

5.7.3.2. CallbackUrl format for Custom Commissioning Flow response

If a **CallbackUrl** field (i.e. **MTcu=**) query field placeholder is present in the **CommissioningCustomFlowUrl**, the Commissioner MAY replace the placeholder value "_" in the **ExpandedCommissioningCustomFlowUrl** with a URL that the manufacturer custom flow can use to make a smooth return to the Commissioner when the device is in a state that it can be commissioned.

This URL field MAY contain a query component (see [RFC 3986](#) section 3.4).

If a query is present, it SHALL be composed of one or more key-value pairs:

- The query SHALL use the **&** delimiter between key/value pairs.
- The key-value pairs SHALL follow the format **name=<value>** where **name** is the key name, and **<value>** is the contents of the value encoded with proper URL-encoded escaping.
- If key **MTrop** is present, it SHALL have a value of "_" (i.e. **MTrop=_**). This is the placeholder for a "returned onboarding payload" provided by the manufacturer flow to the Commissioner.
- Any key whose name begins with **MT** not mentioned in the previous bullets SHALL be reserved for future use by this specification.

Any other element in the **CallbackUrl** query field not covered by the above rules, as well as the fragment field (if present), SHALL remain as provided by the Commissioner through embedding within the **ExpandedCommissioningCustomFlowUrl**, including the order of query key/value pairs present.

5.7.3.2.1. Expansion of CallbackUrl by the manufacturer custom flow

Once the **CallbackUrl** is obtained by the manufacturer flow, it MAY be expanded to form a final **ExpandedCallbackUrl** URL to be used by proceeding with the following substitution algorithm on the provided **CallbackUrl**:

- If key **MTrop** is present, the manufacturer custom flow having received the initial query containing the **CallbackUrl** MAY compute an **Onboarding Payload** in QR code format including **MT:** prefix, and substitute the placeholder value "_" (i.e. in **MTrop=_**) in the **CallbackUrl** with the desired contents, encoded with proper URL-encoded escaping (see [RFC 3986](#) section 2).
 - The contents of the **MTrop=_** key/value pair in the **ExpandedCallbackUrl** SHALL only be expanded if the manufacturer custom flow, having received the initial query containing the **CallbackUrl**, supports opening a commissioning window on the target device and supports conveying the corresponding onboarding payload to the Commissioner.
 - The return onboarding payload, if provided, SHALL contain an ephemeral **Passcode** and not a permanent code that can be used in a subsequent commissioning window. If the manufacturer wants the **Passcode** embedded in the Onboarding Payload placed on-device or in packaging to be the one used for session establishment with the Commissioner, then the manufacturer SHALL NOT include the **MTop** key in its **CommissioningCustomFlowUrl** and SHALL NOT populate the **MTrop** value in the **CallbackUrl** expansion.
 - The contents of the return onboarding payload, if provided, SHALL be constructed to match the state of the device at the moment the **ExpandedCallbackUrl** is opened. At least one ingredient which needs to be adapted relative to the received Onboarding Payload is the **Custom Flow** field which needs to be **0** for the return onboarding payload.